

Quantitative data: shape, measures of central tendency, and spread



Overview

Review

Graphing the shape: histograms and outliers

Measures of the central tendency: mean and median

Measure of spread: The variance and standard deviation

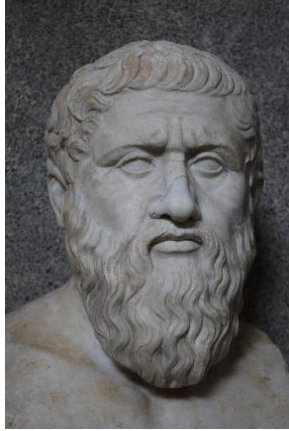
If there is time:

- Normal distributions and z-scores
- Percentile ranks

Review of...

Quantitative variables

Underlying concepts: the P's and the S's



P-Truth

Population or process

Parameter

Plato (Greek symbols)



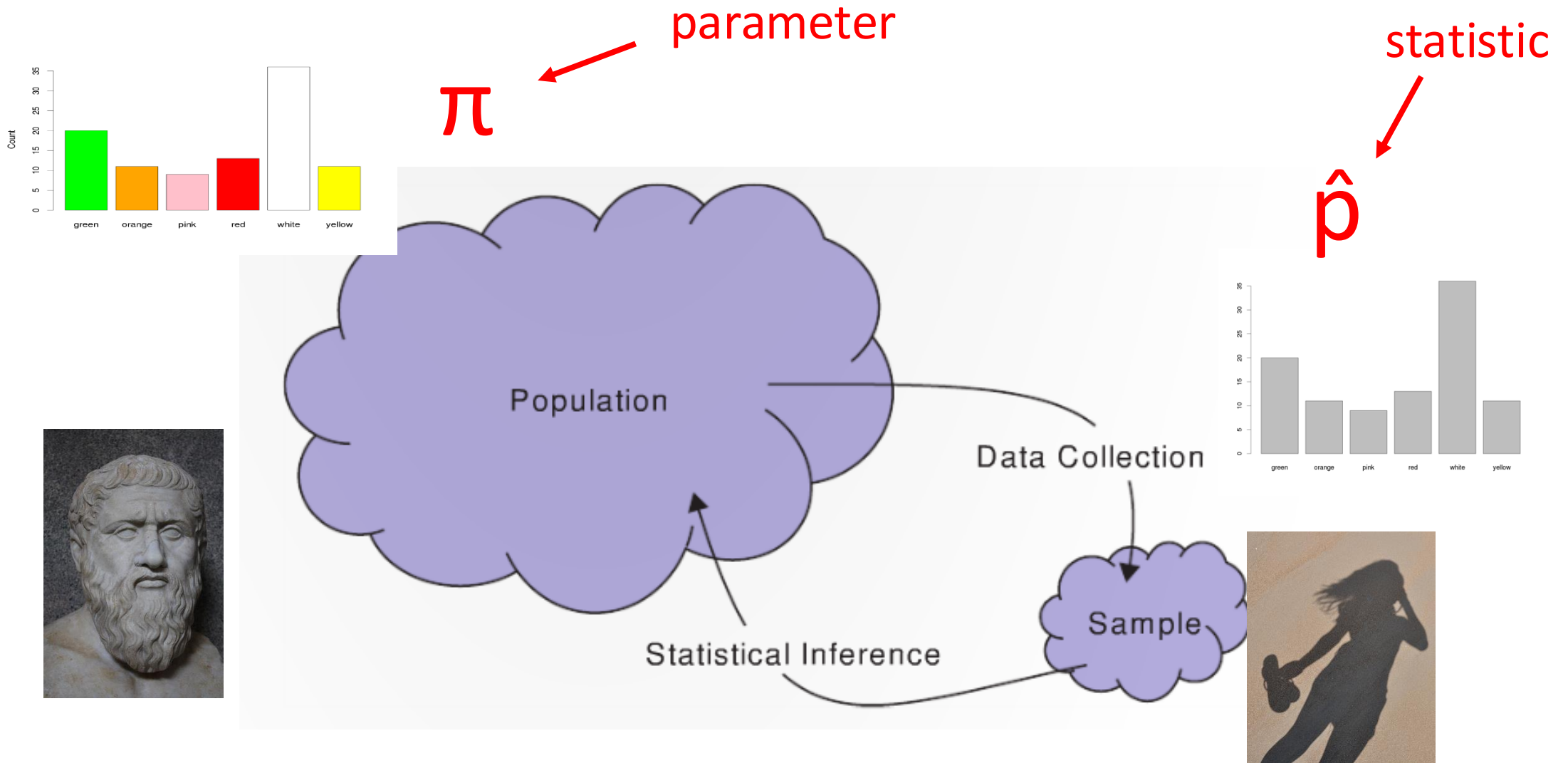
S-shadows

Sample

Statistic

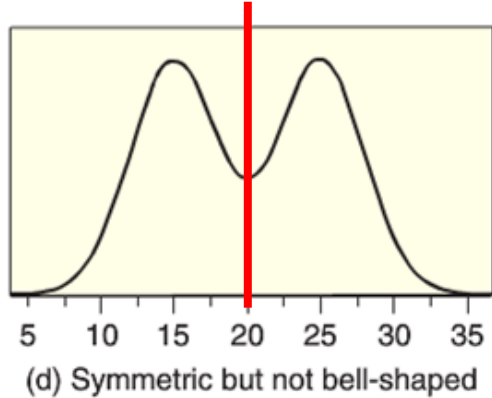
Shadow (Latin symbols)

Review: Categorical data and proportions



Review: Quantitative data and the mean

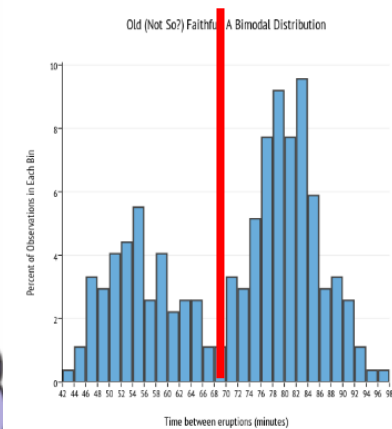
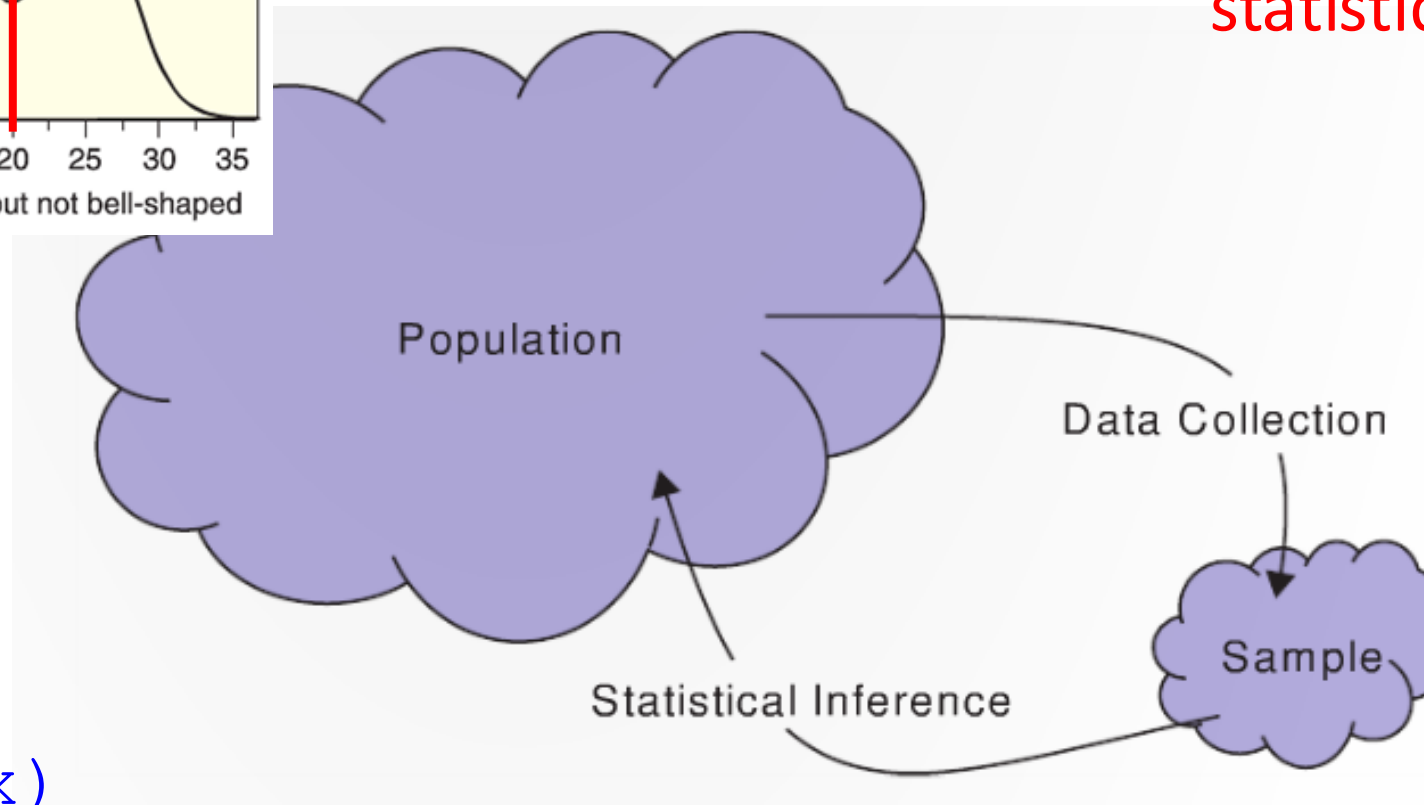
μ ← parameter



$$\bar{x} = \frac{1}{n} \sum_i^n x_i$$

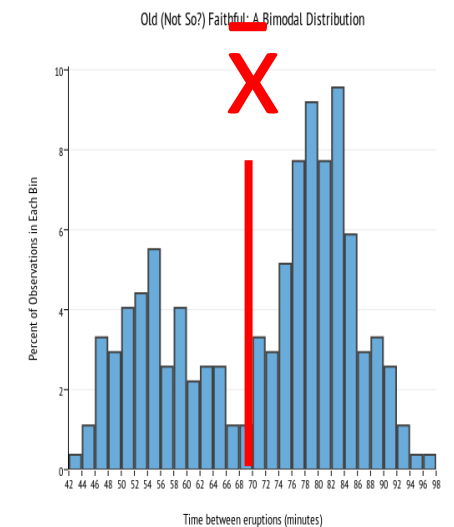
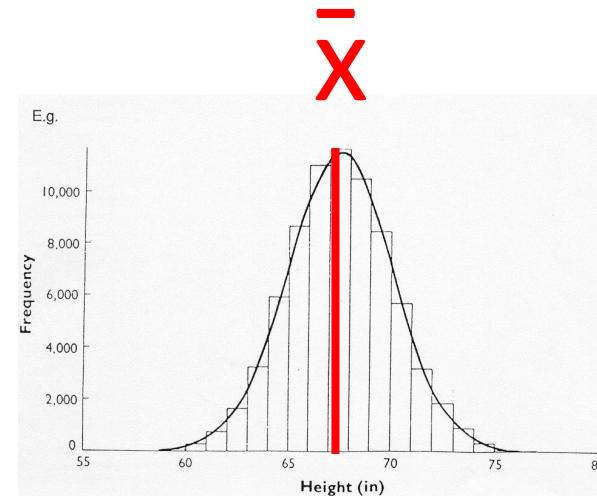
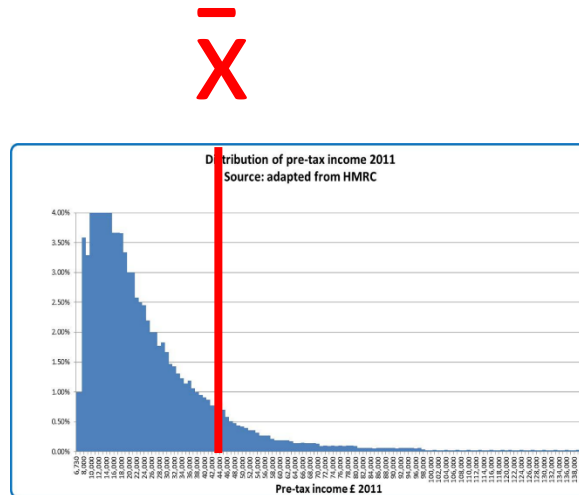
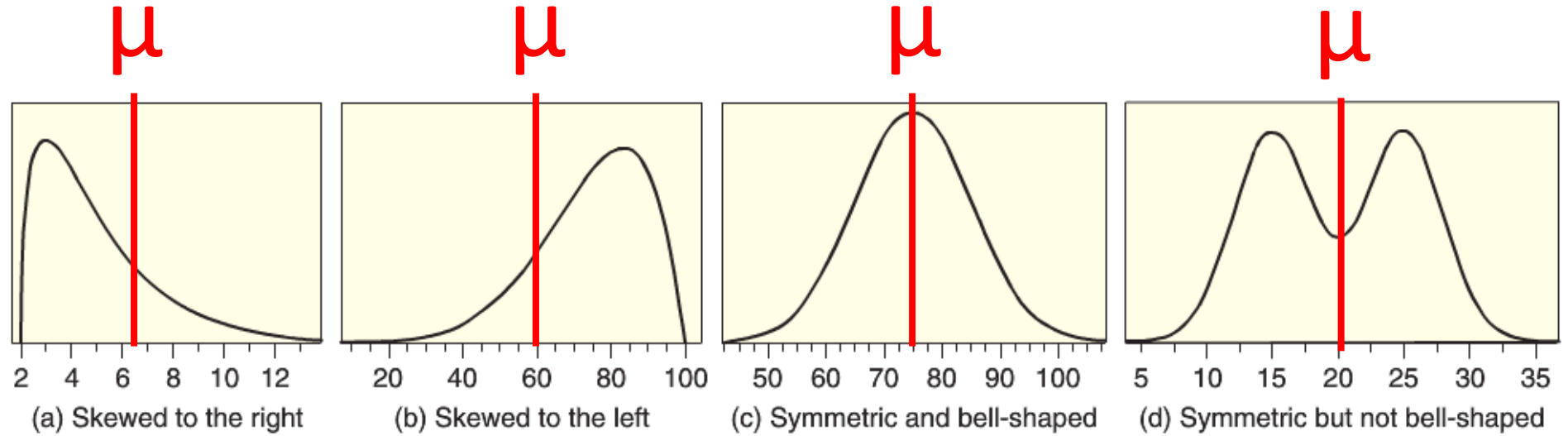
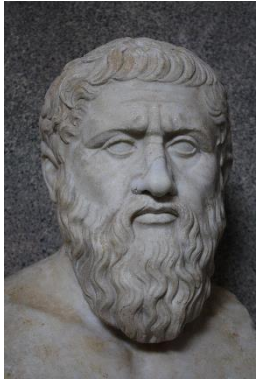
statistic

$\bar{x}$



R: `mean(x)`

Means for differently shaped distributions



Back to the Gapminder data...

get a data frame with information about the countries in the world

```
load("gapminder_2007.Rda")
```

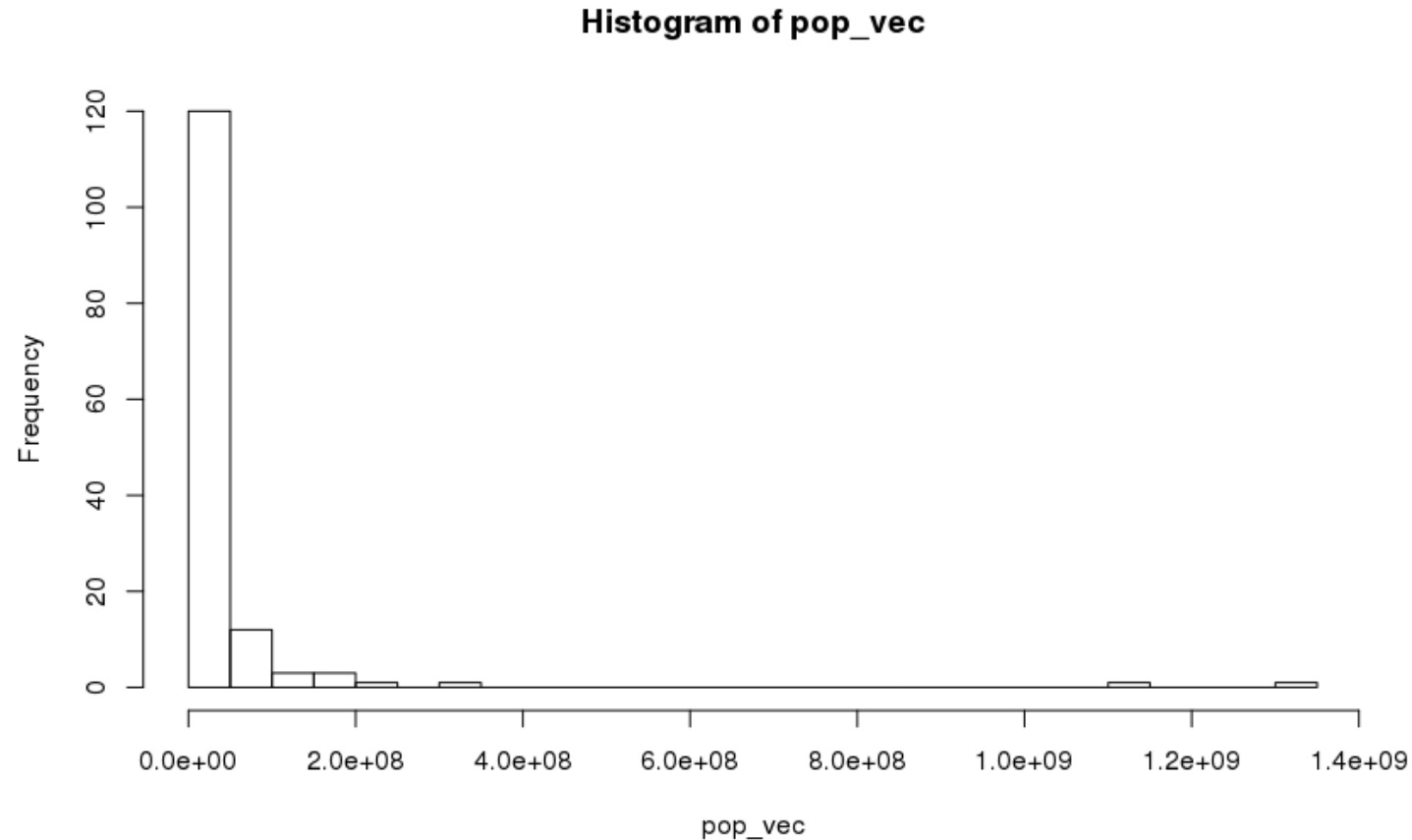
	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	2007	43.828	31889923	974.5803
2	Albania	Europe	2007	76.423	3600523	5937.0295
3	Algeria	Africa	2007	72.301	33333216	6223.3675
4	Angola	Africa	2007	42.731	12420476	4797.2313
5	Argentina	Americas	2007	75.320	40301927	12779.3796

Can you plot a histogram of the population of each country with 20 bins?

```
pop_vec <- gapminder_2007$pop # create a vector with the pop of each country
```

```
hist(pop_vec, breaks = 20) # then create the histogram
```

What is missing from this histogram?



Axes labels could be more informative!

Labeling axes

Question: Can you figure out how to label the axes?

? hist

Answer: xlab and ylab!

```
hist(pop_vec, breaks = 20,  
      ylab = "Frequency",  
      xlab = "Population",  
      main = "World countries population in 2007")
```

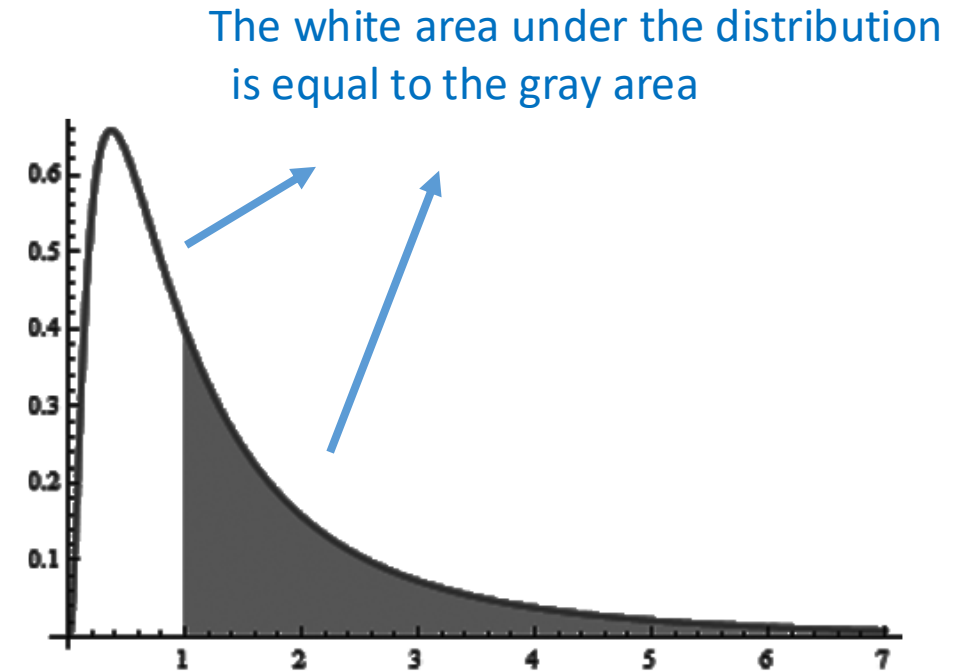
Review: The median

The **median** is a value that splits the data in half

- i.e., half the values in the data are smaller than the median and half are larger

To calculate the median for a data sample of size n , sort the data and then:

- If n is odd: The middle value of the sorted data
- If n is even: The average of the middle two values of the sorted data



R: `median(v)`
`median(v, na.rm = TRUE)`

Review: outliers

Q: What is an **outlier**?

- A: An observed value that is notably distinct from the other values in a dataset

Q: Why are they problematic?

- A: can potentially have a large influence on the statistics we calculate

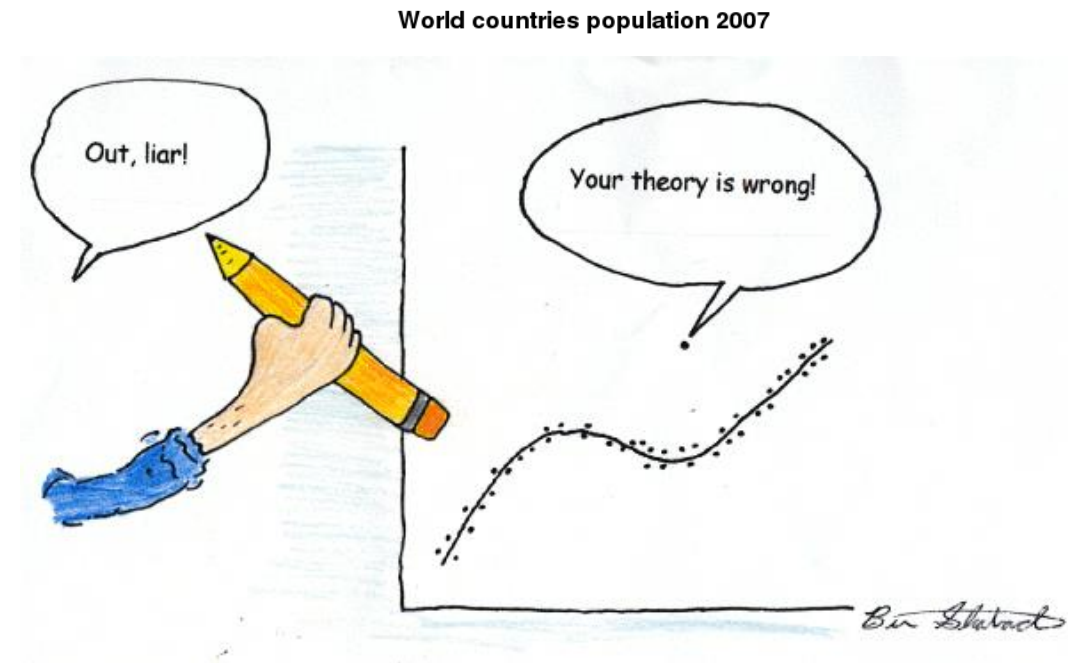
Q: What should you do if you have an outlier in your data?

A: See if you can understand what is causing it!

- If it's an error, delete the point
- If it's a real value, make sure it is not having a big effect on your conclusions, and/or use resistant statistics

Q: Is the mean and/or median resistant?

- A: The median is resistant while the mean is not



Example of calculating the mean and median

When an individual visits a webpage a 'ping' is generated

Below is a random sample of ping counts from 7 people who pinged a website at least once:

4, 6, 10, 12, 45, 59, 158

Question: What is the mean and median ping count in this sample?

A: mean = 42
median = 12

$$\bar{x} = \frac{1}{n} \sum_i^n x_i$$

$$1/7 \cdot 294 = 42$$

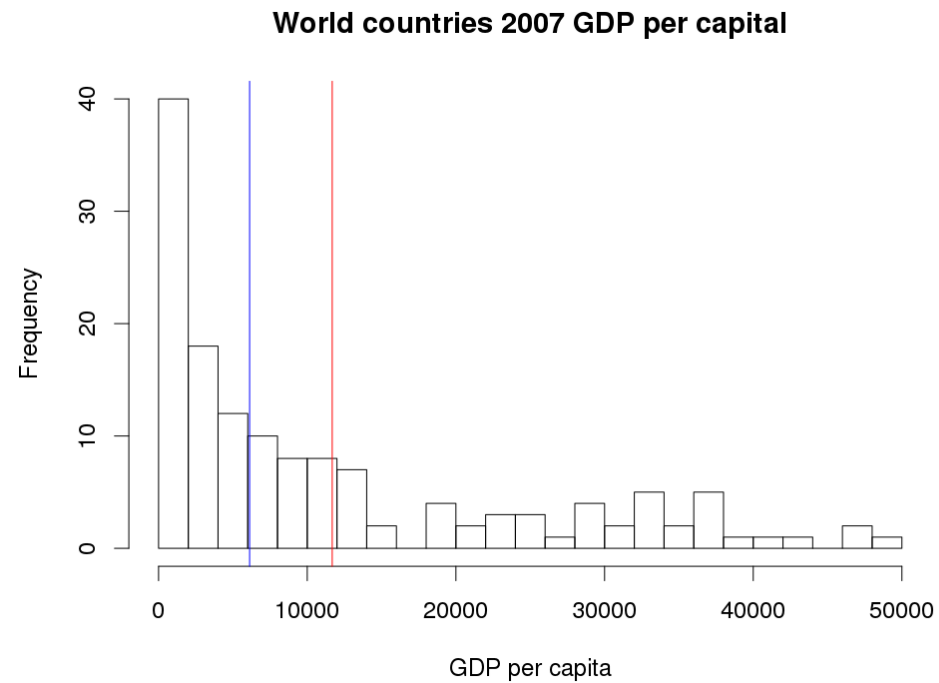


Measures of spread



Characterizing the spread

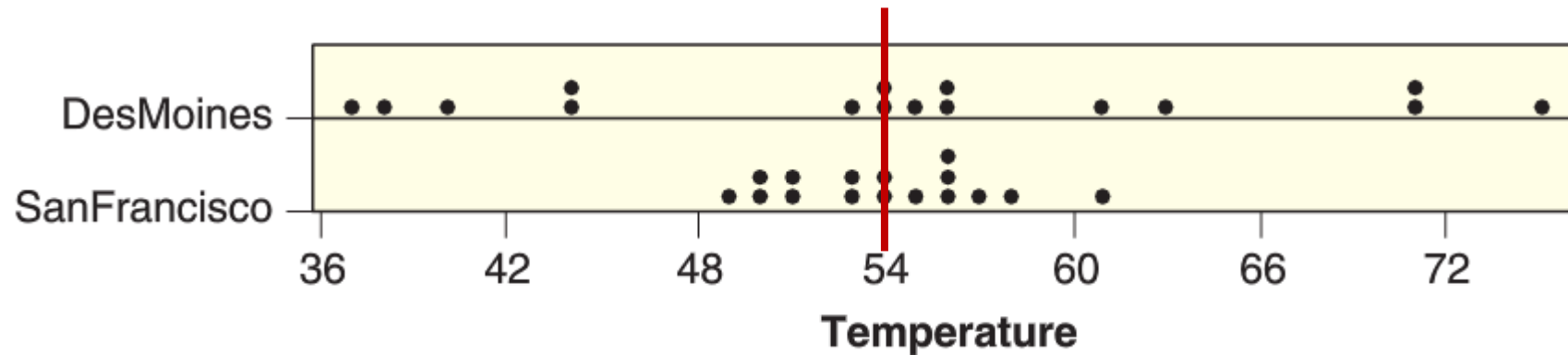
The mean and median are numbers that tell us about the center of a distribution



We can also use numbers to characterize how data is spread

Average monthly temperature: Des Moines vs. San Francisco

Data measured on April 14th from 1997 to 2010:

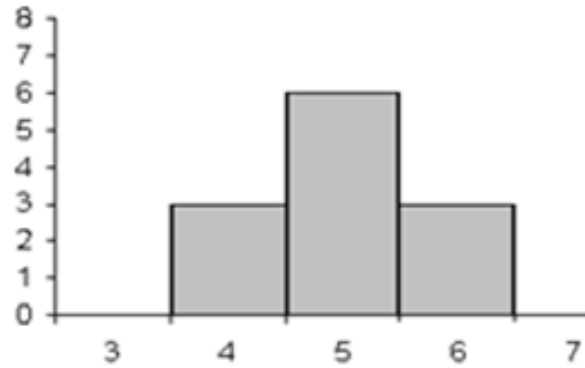


Mean temperature (°F): Des Moines = 54.49 San Fran = 54.01

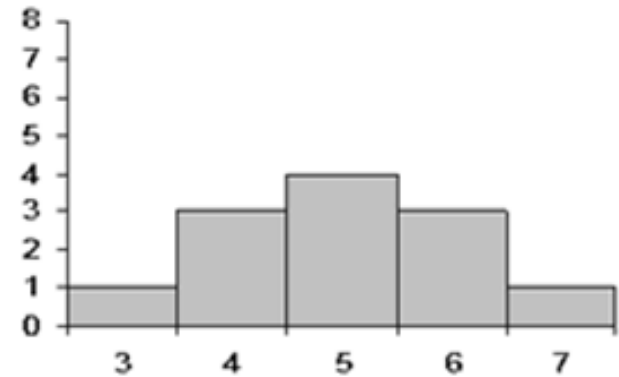
The standard deviation and variance

The **standard deviation** measures the spread of the data

Smaller standard deviation



Larger standard deviation



It gives a rough estimate for a typical distance a point is from the center

The **variance** is the standard deviation squared

Notation

The standard deviation of the ***population*** is denoted σ

- It measure the spread of the data from the population mean μ

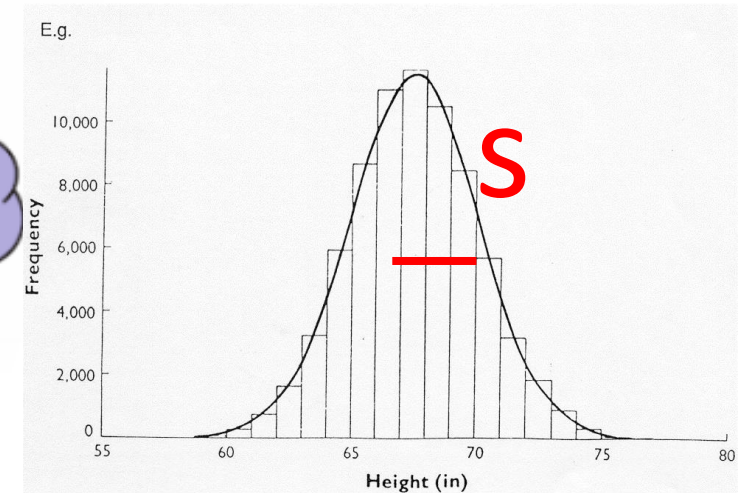
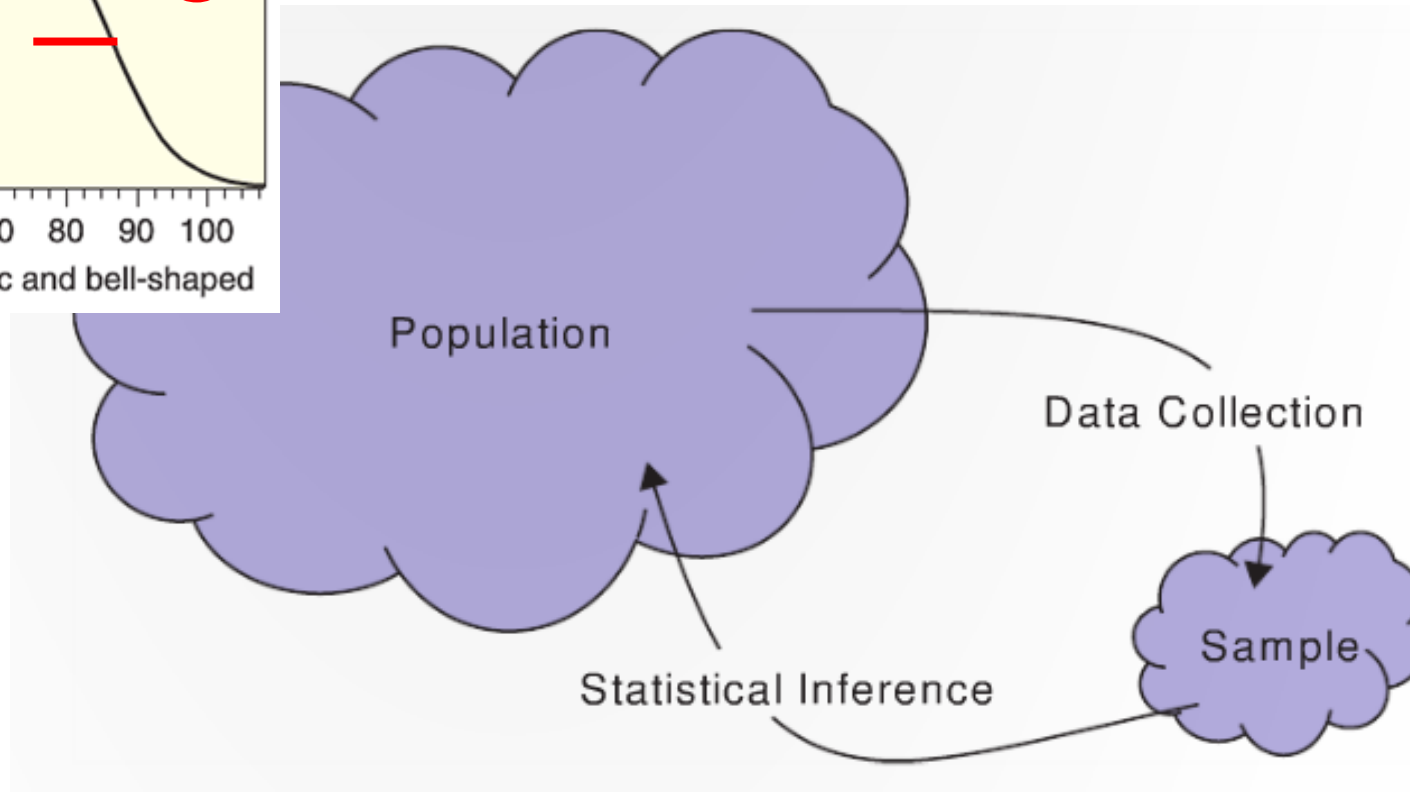
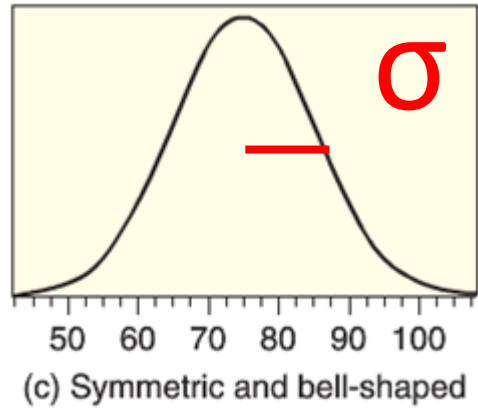
The standard deviation of a ***sample*** is denoted s

- It measure the spread of the data from the sample mean $\bar{x}$

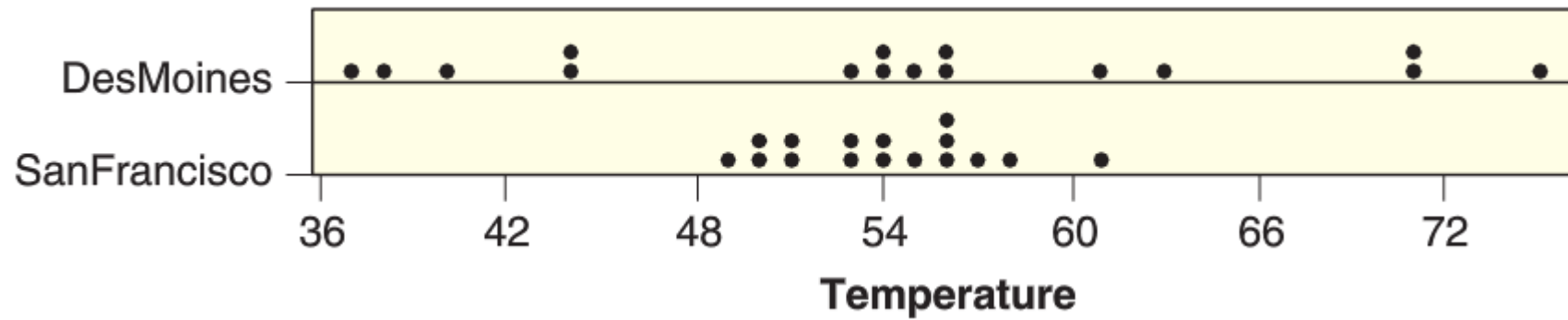
For the variance we use the notation:

- Population variance: σ^2
- Sample variance: s^2

Population and sample standard deviation



Which has the larger standard deviation?



$$s_{DM} = 11.73 \text{ }^{\circ}\text{F}$$

$$s_{SF} = 3.38 \text{ }^{\circ}\text{F}$$

The variance and standard deviation

The **variance** can be computed using the formula:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad \text{R: } \text{var}(v)$$

The **standard deviation** can be computed using the following formula:

- i.e., the standard deviation is the square root of the variance

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{R: } \text{sd}(v)$$

Example: computing the variance

Suppose we had a sample with $n = 4$ points:

$$x_1 = 8, \quad x_2 = 2, \quad x_3 = 6, \quad x_4 = 4$$

We can compute the mean using the formula:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{4} \cdot (x_1 + x_2 + x_3 + x_4) = \frac{1}{4} \cdot (8 + 2 + 6 + 4) = 5$$

The variance can be computed using the formula:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{3} \cdot ((8-5)^2 + (2-5)^2 + (6-5)^2 + (4-5)^2) = 20/3$$

Hot dogs!

Every 4th of July, Nathan's Famous in NYC holds a hot dog eating contest where contestants try to eat as many hot dogs as they can in 10 minutes



$$s = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Homework 2: Calculate the standard deviation “by hand” for the number of hot dogs eaten by the winners.

Let's calculating some variances and standard deviations in R!

Summary of concepts

1. A **probability distribution** shows the **relative likelihood** that we will get a data point in the population with a particular value

- (for a more precise definition take a class in probability)

2. Distributions can have different shapes

- E.g., left skewed, right skewed, bell shaped, etc.

3. The **mean** is one measure of central tendency

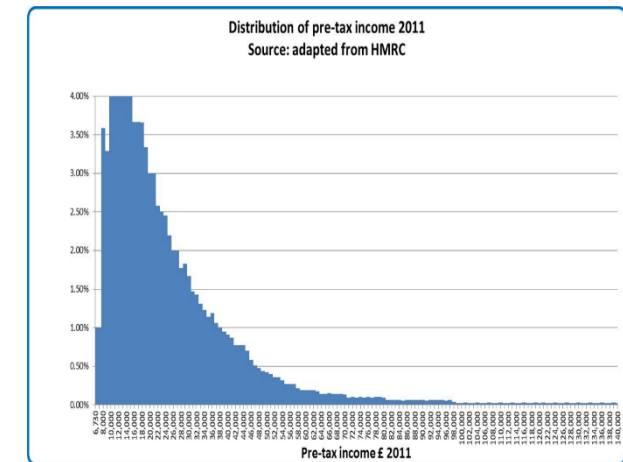
- Sample mean is denoted $\bar{x}$ (statistic)
- Population mean is denoted μ (parameter)

4. The **median** is another measure of central tendency

- The median is resistant to outliers while the mean is not

5. Two measures of variability are the **variance** (σ^2 , s^2) and the **standard deviation** (σ , s)

Income distribution



Summary of R

Data frames contain structured data

- We can extract vectors from a data frame using:

```
my_vec <- my_data_frame$my_var
```

We can get a sense of how quantitative data is distributed by creating a histogram

```
hist(my_vec)
```

We can calculate measures of central tendency using:

```
mean(my_vec)
```

```
median(my_vec)
```

We can calculate measures of variability using:

```
var(my_vec)
```

```
sd(my_vec)
```

Normal distributions and z-scores

z-scores

The z-scores tells how many standard deviations a value is from the mean

- i.e., how far away a point x_i is from $\bar{x}$ in a way that is independent of the units of measurement

$$\text{z-score}(x_i) = \frac{x_i - \bar{x}}{s}$$

Which Accomplishment is most impressive?

LeBron James is a basketball player who had the following statistics in 2011:

- Field goal percentage (FGPct) = 0.510
- Points scored = 2111
- Assists = 554
- Steals = 124



The summary statistics of the NBA in 2011 are given below

$$\text{z-score}(x_i) = \frac{x_i - \bar{x}}{s}$$

	Mean	Standard Deviation
FGPct	0.464	0.053
Points	994	414
Assists	220	170
Steals	68.2	31.5

Try it in R!

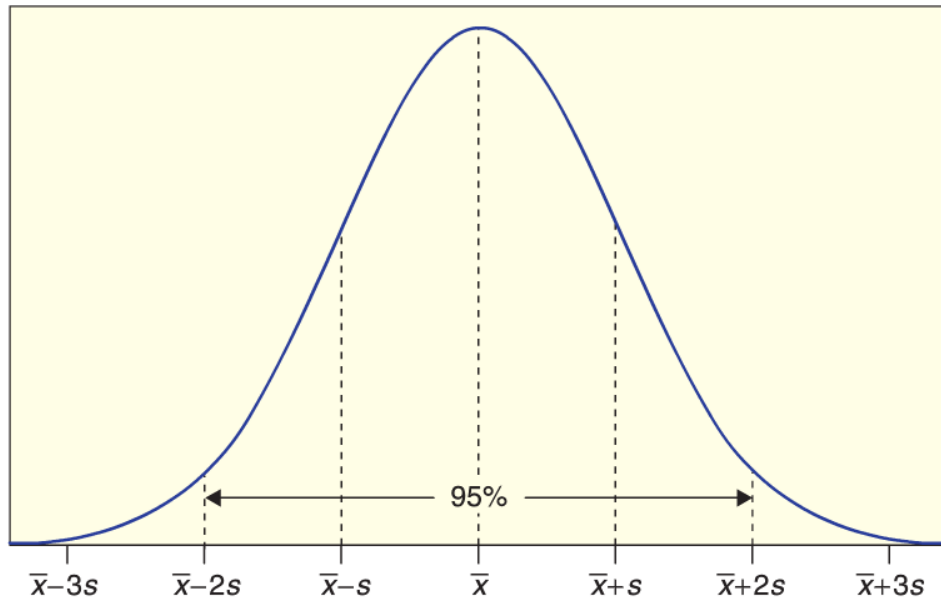
Question: Relative to his peers, which statistic is most and least impressive?

The 95% rule for *normal distributions*

A **normal distribution** is a common distribution that is symmetric and bell shaped

If a distribution of data is approximately normally distributed, about 95% of the data should fall within two standard deviations of the mean

i.e., 95% of the data is in the interval: $\bar{x} - 2s$ to $\bar{x} + 2s$



If points are normally distributed,
should we be impressed with
LeBron's z-score of 2.7?

The 95% rule for *normal distributions*

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Example: IQ scores are normally distributed with a mean of 100 and a standard deviation of 15

Question: what is the range of values that the middle 95% of IQ scores fall in?

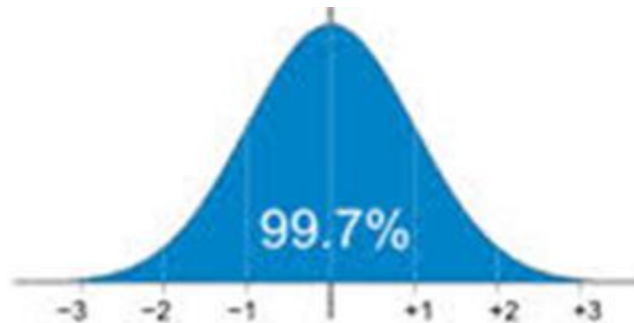
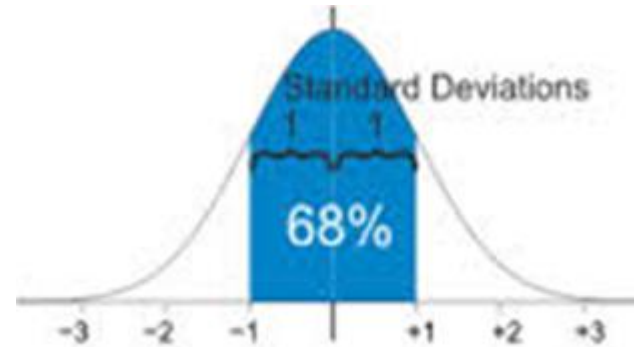
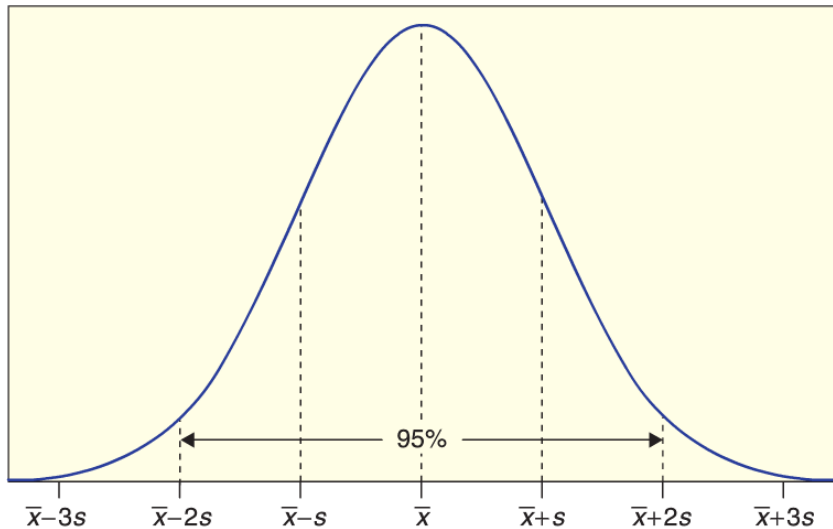
Answer: $(100 - 30)$ to $(100 + 30)$

95% of IQ scores are in the range 70 to 130

The 68%, 95% and 99.7% rules for *normal distributions*

Properties of normal distributions are:

- 68% of the data falls within **one** standard deviations of the mean
- 95% of the data falls within **two** standard deviations of the mean
- 99.7% of the data falls within **three** standard deviations of the mean



Side note: Chebyshev's Inequality

Chebyshev's Inequality: No matter what the shape of the distribution, the proportion of values in the range "average $\pm z \cdot \text{SDs}$ " is at least $1 - 1/z^2$

Range	Proportion
Average ± 2 SDs	at least $1 - 1/4$ (75%)
Average ± 3 SDs	at least $1 - 1/9$ (88.88...%)
Average ± 4 SDs	at least $1 - 1/16$ (93.75%)
Average ± 5 SDs	at least $1 - 1/25$ (96%)

Percentile rank

Percentile rank

The **percentile rank** is the percentage of values in a data set that are less than or equal to a value **x**

We can use `ptail(x, vec_values)` to calculate the percentile rank (returned as a proportion)

For example, suppose we had:

```
ping_data <- c(6, 12, 45, 59, 158)
```

What is the percentile rank of 15?

Answer: 40% `ptail(15, ping_data)` # returns 0.4

We can also get the proportion of values greater or equal to x using:

```
ptail(x, vec_values, lower.tail = FALSE)
```

Let's try it in R!